First Semester B.Tech Degree Regular and Supplementary Examination December 2020.

MAT101

00MAT101121802 -A	Linear Algebra and Calculus

Part A

1) Determine the rank of matrix $A = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & 2 \\ 2 & 4 & -3 \end{bmatrix}$

$R_3 \to R_3 - R_1$ $\begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & 2 \\ 0 & 3 & -2 \end{bmatrix}$

$R_3 \to R_3 - R_2$ $\begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & 2 \\ 0 & 0 & -4 \end{bmatrix}$

$\therefore$ Rank $= 3$

2) Show that the quadratic form $4x^2 + 12xy + 13y^2$ is positive definite.

$$A = \begin{bmatrix} 4 & 6 \\ 6 & 13 \end{bmatrix}$$

characteristic equation is $\lambda^2 - 17\lambda + 16 = 0$

$$(\lambda - 1)(\lambda - 16) = 0$$

$$\lambda = 1, 16$$

Eigen values are positive. Hence the quadratic form is positive definite.

3) If $z = \sin(y^2 - 4x)$, find the rate of change of z with respect to x, at the point $(3,1)$ with y held fixed.

$$\frac{\partial z}{\partial x} = \cos(y^2 - 4x) \times -4$$

$$\left(\frac{\partial z}{\partial x}\right)_{(3,1)} = \cos(1^2 - 4 \times 3) \times -4$$

$$= -4\cos(-11)$$

$$= \underline{\underline{-4\cos(11)}}$$

4) Find $\frac{dz}{dt}$ by chain rule, where $z = 3x^2y^2$, $x = t^4$, $y = t^3$.

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$$

$$\frac{\partial z}{\partial x} = 3y^2 \cdot 2x = 6xy^2 = 6t^4 (t^3)^2 = 6t^4 \cdot t^6 = 6t^{10}$$

$$\frac{\partial z}{\partial y} = 3x^2 \cdot 2y = 6x^2 y = 6(t^4)^2 t^3 = 6t^8 \cdot t^3 = 6t^{11}$$

$$\therefore \frac{dz}{dt} = 6t^{10} \cdot 3t^3 + 6t^{11} \cdot 3t^2$$

$$= 18t^{13} + 18t^{13} = \underline{\underline{36t^{13}}}$$

5) Find the mass of the lamina with density function x^2, which is bounded by $y = x$ and $y = x^2$

Mass of the lamina $= \iint_R \delta(x,y)\, dA$

Here $\delta(x,y) = x^2$

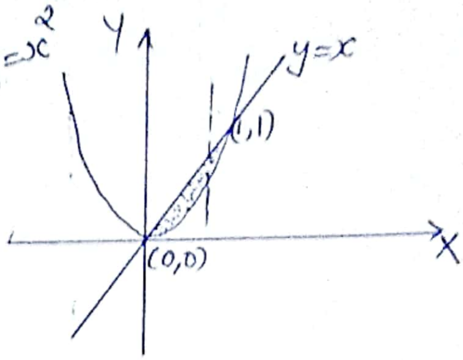
$$\text{Mass} = \int_0^1 \int_{x^2}^{x} x^2 \, dy\, dx$$

$$= \int_0^1 x^2 dx \cdot [y]_{x^2}^{x}$$

$$= \int_0^1 x^2 (x - x^2) dx$$

$$= \int_0^1 (x^3 - x^4) dx = \left[\frac{x^4}{4} - \frac{x^5}{5}\right]_0^1 = \frac{1}{4} - \frac{1}{5}$$

$$= \frac{1}{20}$$



6) Evaluate $\iint_R y^2 x \, dA$ over the region $R = \{(x,y), -3 \le x \le 2, 0 \le y \le 1\}$

$$\iint_R y^2 x \, dA = \int_{-3}^{2}\int_0^1 y^2 x \, dy\, dx = \int_{-3}^{2} x\, dx \left[\frac{y^3}{3}\right]_0^1$$

$$= \frac{1}{3}\int_{-3}^{2} x\, dx = \frac{1}{3}\left[\frac{x^2}{2}\right]_{-3}^{2}$$

$$= \frac{1}{6}(4 - 9) = \frac{-5}{6}$$

7) Test the convergence of the series $\sum_{k=1}^{\infty} \left(\frac{k}{100}\right)^k$

$u_k = \left(\frac{k}{100}\right)^k$. Then $(u_k)^{1/k} = \frac{k}{100}$

$$\lim_{k\to\infty} (u_k)^{1/k} = \lim_{k\to\infty} \frac{k}{100} = \infty > 1$$

$\therefore$ By Root test, the geven series diverges.

8) Does the series $\sum_{k=1}^{\infty}\left(\frac{-3}{4}\right)^k$ converge? If so, find the sum.

$\sum_{k=1}^{\infty}\left(\frac{-3}{4}\right)^k$ is a geometric series, with common ratio $r = \frac{-3}{4}$.

Here $|r| = \left|\frac{-3}{4}\right| = \frac{3}{4} < 1$.

So the geometric series converges.

$$\text{Sum} = \frac{a}{1-r} = \frac{-3/4}{1-\frac{-3}{4}} = \frac{-3/4}{7/4}$$

$$= \frac{-3}{7}$$

9) Find the binomial series for $f(x) = (1+x)^{1/3}$ up to third degree term.

$$(1+x)^m = 1 + mx + \frac{m(m-1)}{2!}x^2 + \frac{m(m-1)(m-2)}{3!}x^3 + \cdots$$

$$(1+x)^{1/3} = 1 + \frac{1}{3}x + \frac{\frac{1}{3}(\frac{1}{3}-1)}{2!}x^2 + \frac{\frac{1}{3}(\frac{1}{3}-1)(\frac{1}{3}-2)}{3!}x^3 + \cdots$$

$$= 1 + \frac{1}{3}x + \frac{\frac{1}{3}\cdot\frac{-2}{3}}{2}x^2 + \frac{\frac{1}{3}\cdot\frac{-2}{3}\cdot\frac{-5}{3}}{6}x^3 + \cdots$$

$$= 1 + \frac{1}{3}x - \frac{1}{9}x^2 + \frac{5}{81}x^3 + \cdots$$

10) Find the Maclaurin's series of $f(x) = \log(1+x)$ up to third degree term.

Maclaurin series

$$f(x) = f(0) + \frac{x}{1!}f'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \cdots$$

$f(x) = \log(1+x)$, $f(0) = \log 1 = 0$

$f'(x) = \frac{1}{1+x}$, $f'(0) = 1$

$= (1+x)^{-1}$

$f''(x) = -1(1+x)^{-2}$, $f''(0) = -1$

$f'''(x) = 2(1+x)^{-3}$, $f'''(0) = 2$

$\therefore \log(1+x) = 0 + \frac{x}{1!} \times 1 + \frac{x^2}{2!} \times -1 + \frac{x^3}{3!} \times 2 + \cdots$

$= \frac{x}{1!} - \frac{x^2}{2!} + \frac{2x^3}{3!} \cdots$

PART – B

Module – 1

11

a) Solve the following linear system of equations using Gauss elimination method.

$x + y + z = 6$, $x + 2y - 3z = -4$, $-x - 4y + 9z = 18$

$$\tilde{A} = \begin{bmatrix} 1 & 1 & 1 & 6 \\ 1 & 2 & -3 & -4 \\ -1 & -4 & 9 & 18 \end{bmatrix}$$

$R_2 \to R_2 - R_1$, $R_3 \to R_3 + R_1$

$$\begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & -4 & -10 \\ 0 & -3 & 10 & 24 \end{bmatrix}$$

$R_3 \to R_3 + 3R_2$

$$\begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & -4 & -10 \\ 0 & 0 & -2 & -6 \end{bmatrix}$$

$-2z = -6 \qquad \therefore \underline{\underline{z = 3}}$

$y - 4z = -10$

$y - 4 \times 3 = -10 \qquad \therefore y = -10 + 12$

$\qquad\qquad\qquad\qquad\qquad = \underline{\underline{2}}$

$x + y + z = 6$

$x + 2 + 3 = 6 \qquad \therefore \underline{\underline{x = 1}}$

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b) Find eigen values and eigen vectors of the matrix

$$A = \begin{bmatrix} 1 & 1 & 2 \\ -1 & 2 & 1 \\ 0 & 1 & 3 \end{bmatrix}$$

Characteristic equation is $|A - \lambda I| = 0$

$$\lambda^3 - \lambda^2(\text{Trace of } A) + \lambda\,(\text{Sum of minors of diagonal elements}) - |A| = 0$$

Trace $= 1 + 2 + 3 = 6$

$M_{11} = \begin{vmatrix} 2 & 1 \\ 1 & 3 \end{vmatrix} = 6 - 1 = 5$

$M_{22} = \begin{vmatrix} +1 & 2 \\ 0 & 3 \end{vmatrix} = +3 - 0 = +3$

$M_{33} = \begin{vmatrix} 1 & 1 \\ -1 & 2 \end{vmatrix} = 2 - -1 = 3$

Sum of minors $= 5 + 3 + 3 = 11$

$|A| = 1(6-1) - 1(-3-0) + 2(-1-0)$

$\quad = 5 + 3 - 2 = 6$

$\therefore \lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$

$\lambda = 1, 2, 3$

Eigen values are $\underline{\underline{1, 2, 3}}$

Eigen Vector when $\lambda = 1$: $(A-I)X=0$

$$\begin{bmatrix} 0 & 1 & 2 \\ -1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$R_1 \leftrightarrow R_2$

$$\begin{bmatrix} -1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix} \Longrightarrow \quad R_3 \to R_3 - R_2 \quad \begin{bmatrix} -1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

Rank is 2 and number of unknowns is 3

So let $x_3 = k$

$$x_2 + 2x_3 = 0 \Rightarrow x_2 + 2k = 0 \Rightarrow x_2 = -2k$$

$$-x_1 + x_2 + x_3 = 0 \Rightarrow -x_1 - 2k + k = 0$$

$$\Rightarrow -x_1 - k = 0$$

$$\Rightarrow x_1 = -k$$

The eigen vector $X_1 = \begin{bmatrix} -k \\ -2k \\ k \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix}$

Eigen vector when $\lambda = 2$: $(A-2I)X = 0$

$$\begin{bmatrix} -1 & 1 & 2 \\ -1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$R_2 \to R_2 - R_1$

$$\begin{bmatrix} -1 & 1 & 2 \\ 0 & -1 & -1 \\ 0 & 1 & 1 \end{bmatrix}$$

$R_3 \to R_3 + R_2$

$$\begin{bmatrix} -1 & 1 & 2 \\ 0 & -1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

Rank is 2 and number of unknown is 3. So let

$x_3 = a$

$-x_2 - x_3 = 0 \Rightarrow -x_2 - a = 0 \Rightarrow x_2 = -a$

$-x_1 + x_2 + 2x_3 = 0 \Rightarrow -x_1 - a + 2a = 0$

$\Rightarrow -x_1 + a = 0$

$\Rightarrow x_1 = a$

Eigen vector $X_2 = \begin{bmatrix} a \\ -a \\ a \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

Eigen vector when $\lambda = 3$: $(A - 3I)X = 0$

$$\begin{bmatrix} -2 & 1 & 2 \\ -1 & -1 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$R_2 \rightarrow -2R_2 + R_1$

$$\begin{bmatrix} -2 & 1 & 2 \\ 0 & 3 & 0 \\ 0 & 1 & 0 \end{bmatrix} \Rightarrow$$

$R_3 \rightarrow 3R_3 - R_2$

$$\begin{bmatrix} -2 & 1 & 2 \\ 0 & 3 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Here $x_3 = b$

$3x_2 = 0 \Rightarrow x_2 = 0$

$-2x_1 + x_2 + 2x_3 = 0$

$-2x_1 + 0 + 2b = 0 \Rightarrow x_1 = b$

Eigen vector $X_3 = \begin{bmatrix} b \\ 0 \\ b \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

12

a) Show that the equations $x+y+z=a$, $3x+4y+5z=b$, $2x+3y+4z=c$ (i) have no solution if $a=b=c=1$ (ii) have many solutions if $a=\frac{b}{2}=c=1$.

$$\tilde{A} = \begin{bmatrix} 1 & 1 & 1 & a \\ 3 & 4 & 5 & b \\ 2 & 3 & 4 & c \end{bmatrix}$$

$R_2 \to R_2 - 3R_1$, $R_3 \to R_3 - 2R_1$

$$\begin{bmatrix} 1 & 1 & 1 & a \\ 0 & 1 & 2 & b-3a \\ 0 & 1 & 2 & c-2a \end{bmatrix}$$

$R_3 \to R_3 - R_2$

$$\begin{bmatrix} 1 & 1 & 1 & a \\ 0 & 1 & 2 & b-3a \\ 0 & 0 & 0 & a-b+c \end{bmatrix}$$

(i)
The system will have no solution if $\text{Rank}[\tilde{A}] \neq \text{Rank}[A]$

ie when $a-b+c \neq 0$.

so if $a=b=c=1$, $a-b+c = 1-1+1 = 1 \neq 0$

Hence <u>no</u> solution.

(ii)
The system will have many solution if

$\text{Rank}[\tilde{A}] = \text{Rank}[A] <$ no. of unknowns.

Here number of unknowns $= 3$.

So $\text{Rank}[\tilde{A}] = \text{Rank}[A]$, should be less than 3.

Rank $[A]=2$. Rank $[\tilde{A}]=2$ if $b-3a\neq 0$ and

$a-b+c=0$.

So when $a=\frac{b}{2}=c=1 \Rightarrow a=1,\ b=2,\ c=1$

$b-3a\neq 0$ and $a-b+c=0$

The system will have infinitely many solutions

b) Find the matrix of transformation that diagonalize the matrix $A=\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$. Also find the diagonal matrix.

characteristic equation is $|A-\lambda I|=0$

$$\lambda^3-\lambda^2(\text{trace of } A)+\lambda\left(\begin{matrix}\text{Sum of minors}\\ \text{of main diagonal}\end{matrix}\right)-|A|=0$$

Trace $=6+3+3=12$

$M_{11}=\begin{vmatrix} 3 & -1 \\ -1 & 3 \end{vmatrix}=9-1=8$

$M_{22}=\begin{vmatrix} +6 & +2 \\ 2 & 3 \end{vmatrix}=\overset{18-4}{\cancel{6-2}}=+14$

$M_{33}=\begin{vmatrix} 6 & -2 \\ -2 & 3 \end{vmatrix}=18-4=14$

Sum of minors $=8+14+14=\cancel{18}\,36$

$|A|=6(9-1)+2(-6+2)+2(2-6)$

$=48-8-8=\cancel{48}\,32$

$\lambda^3-12\lambda^2+\overset{36}{\cancel{18}}\lambda-\overset{32}{\cancel{48}}=0$

$\lambda=\underline{2,2,8}$

Eigen vector when $\lambda = 2$

$(A - 2I)X = 0$

$$\begin{bmatrix} 4 & -2 & 2 \\ -2 & 1 & -1 \\ 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$R_2 \to 2R_2 + R_1$, $R_3 \to 2R_3 - R_1$

$$\begin{bmatrix} 4 & -2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Rank = 1 , But number of unknowns = 3

So we may have to introduce two arbitrary constants.

Let $x_2 = a$, $x_3 = b$

Then $4x_1 - 2x_2 + 2x_3 = 0$

$4x_1 - 2a + 2b = 0$

$4x_1 = 2a - 2b$

$x_1 = \frac{2(a-b)}{4} = \frac{a-b}{2}$

Eigen vector

$$X = \begin{bmatrix} \frac{a-b}{2} \\ a \\ b \end{bmatrix} = \begin{bmatrix} \frac{a}{2} \\ a \\ 0 \end{bmatrix} + \begin{bmatrix} -b/2 \\ 0 \\ b \end{bmatrix}$$

$$X_1 = \begin{bmatrix} 1/2 \\ 1 \\ 0 \end{bmatrix} = \underline{\underline{\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}}} \quad \text{and} \quad X_2 = \begin{bmatrix} -1/2 \\ 0 \\ 1 \end{bmatrix} = \underline{\underline{\begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}}}$$

Eigen vector when $\lambda = 8$

$(A - 8I)X = 0$

$$\begin{bmatrix} -2 & -2 & 2 \\ -2 & -5 & -1 \\ 2 & -1 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$R_2 \to R_2 - R_1$, $R_3 \to R_3 + R_1$

$$\begin{bmatrix} -2 & -2 & 2 \\ 0 & -3 & -3 \\ 0 & -3 & -3 \end{bmatrix}$$

$R_3 \to R_3 - R_2 \Rightarrow \begin{bmatrix} -2 & -2 & 2 \\ 0 & -3 & -3 \\ 0 & 0 & 0 \end{bmatrix}$

Rank is 2 and number of unknowns is 3. So let

$x_3 = k$

$-3x_2 - 3x_3 = 0 \Rightarrow -3x_2 - 3k = 0$

$\Rightarrow x_2 = -k$

$-2x_1 - 2x_2 + 2x_3 = 0 \Rightarrow -2x_1 + 2k + 2k = 0$

$\Rightarrow x_1 = 2k$

Eigen vector $X_3 = \begin{bmatrix} 2k \\ -k \\ k \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$

Then $B = \begin{bmatrix} 1 & -1 & 2 \\ 2 & 0 & -1 \\ 0 & 2 & 1 \end{bmatrix}$

$$B^{-1} = \frac{\text{adj } B}{|B|}$$

$|B| = 1(0+2) + 1(2-0) + 2(4-0)$

$= 2 + 2 + 8 = 12$

$c_{11} = 2$ $\quad c_{21} = -(-5) = 5$ $\quad c_{31} = 1$

$c_{12} = -2$ $\quad c_{22} = 1$ $\quad c_{32} = -(-5) = 5$

$c_{13} = 4$ $\quad c_{23} = -2$ $\quad c_{33} = 2$

$$B^{-1} = \frac{\begin{bmatrix} 2 & 5 & 1 \\ -2 & 1 & 5 \\ 4 & -2 & 2 \end{bmatrix}}{12}$$

Then $B^{-1}AB = \frac{1}{12}\begin{bmatrix} 2 & 5 & 1 \\ -2 & 1 & 5 \\ 4 & -2 & 2 \end{bmatrix}\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}\begin{bmatrix} 1 & -1 & 2 \\ 2 & 0 & -1 \\ 0 & 2 & 1 \end{bmatrix}$

$$= \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 8 \end{bmatrix}$$

Module - 2

13 a) Find the local linear approximation of $\frac{4y}{x+z}$ at $(1,1,1)$.

Let $f(x,y,z) = \frac{4y}{x+z}$, $f(1,1,1) = \frac{4}{2} = 2$

$f_x = 4y \cdot -1(x+z)^{-2} = \frac{-4y}{(x+z)^2}$, $f_x(1,1,1) = \frac{-4}{2^2} = -1$

$f_y = \frac{4}{x+z} \cdot 1$, $f_y(1,1,1) = \frac{4}{2} = 2$

$f_z = 4y \cdot -(x+z)^{-2} = \frac{-4y}{(x+z)^2}$, $f_z(1,1,1) = \frac{-4}{2^2} = -1$

Local linear approximation is

$L(x,y,z) = f(x_0,y_0,z_0) + f_x(x-x_0) + f_y(y-y_0) + f_z(z-z_0)$

$= 2 + -1(x-1) + 2(y-1) + -(z-1)$

$= 2 - x + 1 + 2y - 2 - z + 1 = -x + 2y - z + 2$

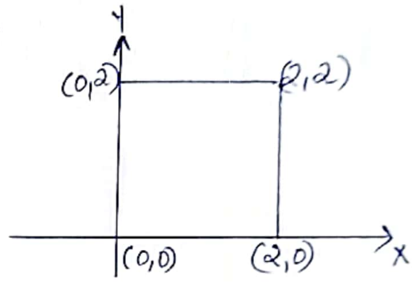
b) Find the absolute extrema of the function $f(x,y) = x^2 - 3y^2 - 2x + 6y$ over the square region with vertices $(0,0)$, $(0,2)$, $(2,2)$ and $(2,0)$.

$$\frac{\partial f}{\partial x} = 2x - 2 \quad , \quad \frac{\partial f}{\partial x} = 0 \Rightarrow 2x - 2 = 0$$

$$\underline{x = 1}$$

$$\frac{\partial f}{\partial y} = -6y + 6 \quad , \quad \frac{\partial f}{\partial y} = 0 \Rightarrow -6y + 6 = 0 \Rightarrow \underline{y = 1}$$

$\therefore \underline{(1,1)}$ is a critical point.



The equation of straight line joining $(0,0)$ and $(2,0)$ is $y = 0$

$$\therefore f(x,0) = x^2 - 2x$$

$$f'(x,0) = 2x - 2$$

$$f'(x,0) = 0 \Rightarrow 2x - 2 = 0 \Rightarrow x = 1$$

$\therefore \underline{(1,0)}$ is a critical point

The equation of straight line joining $(2,0)$ and $(2,2)$ is $x = 2$.

$$f(2,y) = 4 - 3y^2 - 4 + 6y$$

$$f'(2,y) = -6y + 6$$

$$f'(2,y) = 0 \Rightarrow -6y + 6 = 0 \Rightarrow y = 1$$

$\therefore \underline{(2,1)}$ is a critical point

The eqn. of straight line joining $(2,2)$ and $(0,2)$ is $y = 2$.

$$f(x,2) = x^2 - 12 - 2x + 12$$

$$f'(x,2) = 2x - 2$$

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$f'(x,2)=0 \rightarrow 2x-2=0 \Rightarrow x=1$

$\therefore \underline{(1,2)}$ is a critical point

The eqn. of straight line joining (0,2) and (0,0) is $x=0$.

$f(0,y) = -3y^2+6y$

$f'(0,y) = -6y+6$

$f'(0,y)=0 \Rightarrow -6y+6=0 \Rightarrow y=1$

$\therefore \underline{(0,1)}$ is a critical point

(x,y)	(1,1)	(1,0)	(2,1)	(1,2)	(0,1)	(0,0)	(2,0)	(0,2)	(2,2)
f(x,y)	2	-1	3	-1	3	0	0	0	0

Hence the absolute maxima is at the points (2,1) and (0,1). The absolute maximum value is 3.

The absolute minima is at the points (1,0) and (1,2). The absolute minimum value is -1.

14a) If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$ find the value of $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial}{\partial}$

Let $p = \frac{x}{y}$, $q = \frac{y}{z}$, $r = \frac{z}{x}$

Then $u = f(p,q,r)$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial p}\cdot\frac{\partial p}{\partial x} + \frac{\partial u}{\partial q}\cdot\frac{\partial q}{\partial x} + \frac{\partial u}{\partial r}\cdot\frac{\partial r}{\partial x}$$

$$= \frac{\partial u}{\partial p}\cdot\frac{1}{y} + \frac{\partial u}{\partial q}\times 0 + \frac{\partial u}{\partial r}\cdot\frac{-z}{x^2}.$$

$$\frac{\partial u}{\partial x} = \frac{1}{y}\frac{\partial u}{\partial p} - \frac{z}{x^2}\frac{\partial u}{\partial r} \quad \text{——— ①}$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial p}\frac{\partial p}{\partial y} + \frac{\partial u}{\partial q}\frac{\partial q}{\partial y} + \frac{\partial u}{\partial r}\frac{\partial r}{\partial y}$$

$$= \frac{\partial u}{\partial p}\cdot\frac{-x}{y^2} + \frac{\partial u}{\partial q}\cdot\frac{1}{z} + \frac{\partial u}{\partial r}\times 0$$

$$\frac{\partial u}{\partial y} = -\frac{x}{y^2}\frac{\partial u}{\partial p} + \frac{1}{z}\frac{\partial u}{\partial q} \quad \text{——— ②}$$

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial p}\frac{\partial p}{\partial z} + \frac{\partial u}{\partial q}\frac{\partial q}{\partial z} + \frac{\partial u}{\partial r}\cdot\frac{\partial r}{\partial z}$$

$$= \frac{\partial u}{\partial p}\times 0 + \frac{\partial u}{\partial q}\cdot\frac{-y}{z^2} + \frac{\partial u}{\partial r}\cdot\frac{1}{x}$$

$$\frac{\partial u}{\partial z} = \frac{-y}{z^2}\frac{\partial u}{\partial q} + \frac{1}{x}\frac{\partial u}{\partial r} \quad \text{——— ③}$$

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = \frac{x}{y}\frac{\partial u}{\partial p} - \frac{z}{x}\frac{\partial u}{\partial r} - \frac{x}{y}\frac{\partial u}{\partial p}$$

$$+ \frac{\overset{y}{\cancel{x}}}{z}\frac{\partial u}{\partial q} - \frac{y}{z}\frac{\partial u}{\partial q} + \frac{z}{x}\frac{\partial u}{\partial r}$$

$$= 0$$

b) Locate all relative extrema of $f(x,y) = 2xy - x^3 - y^2$

$f_x = 2y - 3x^2$, $f_x = 0 \Rightarrow 2y - 3x^2 = 0$

$f_y = 2x - 2y$, $f_y = 0 \Rightarrow 2x - 2y = 0$

$\Rightarrow x = y$

substitute in $f_x = 0$, then

$2y - 3x^2 = 0 \Rightarrow 2x - 3x^2 = 0$

$x(2 - 3x) = 0$

$x = 0$ or $x = 2/3$

$\therefore (0,0), \left(\frac{2}{3}, \frac{2}{3}\right)$ are the critical points.

$r = f_{xx} = -6x$

$t = f_{yy} = -2$

$s = f_{xy} = 2$

At $(0,0)$, $r = 0$, $t = -2$, $s = 2$

So $rt - s^2 = 0 - 4 = -4 < 0$

$\therefore (0,0)$ is a saddle point

At $\left(\frac{2}{3}, \frac{2}{3}\right)$, $r = -6 \times \frac{2}{3} = -4$, $t = -2$, $s = 2$

So $rt - s^2 = 8 - 4 = 4 > 0$ and $r < 0$.

$\therefore \left(\frac{2}{3}, \frac{2}{3}\right)$ is the point of relative maxima.

The relative maximum value is

$$f\left(\frac{2}{3}, \frac{2}{3}\right) = 2 \cdot \frac{2}{3} \cdot \frac{2}{3} - \left(\frac{2}{3}\right)^3 - \left(\frac{2}{3}\right)^2$$

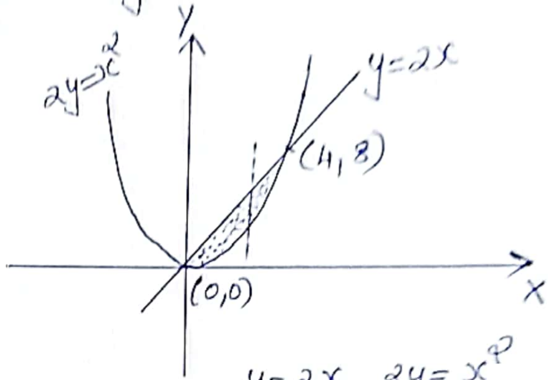
$$= \frac{8}{9} - \frac{8}{27} - \frac{4}{9} = \frac{4}{9} - \frac{8}{27}$$

$$= \frac{12 - 8}{27} = \frac{4}{27}$$

15.

Module - 3

a) Use double integrals to find the area of the region enclosed between the parabola $2y = x^2$ and the line $y = 2x$.



$$\text{Area} = \iint dA$$

$$= \int_0^4 \int_{x^2/2}^{2x} dy\,dx$$

$$= \int_0^4 \left[y\right]_{x^2/2}^{2x} dx = \int_0^4 \left(2x - \frac{x^2}{2}\right) dx$$

$y = 2x$, $2y = x^2$

$2 \times 2x = x^2$

$4x = x^2$

$4 = x$

$$= \left[2 \cdot \frac{x^2}{2} - \frac{1}{2} \cdot \frac{x^3}{3}\right]_0^4$$

$$= \left[x^2 - \frac{x^3}{6}\right]_0^4 = 16 - \frac{64}{6} = 16 - \frac{32}{3}$$

$$= \frac{48 - 32}{3} = \frac{16}{3} \text{ sq. unit}$$

---

b) Find the volume of the solid in the first octant bounded by the coordinate planes and the plane $x + 2y + z = 6$

$$\text{Volume} = \iint_R f(x,y)\,dy\,dx = \iint_R z\,dy\,dx$$

$$= \int_0^6 \int_0^{\frac{6-x}{2}} (6 - x - 2y)\,dy\,dx$$

$$= \int_0^6 \left[6y - xy - y^2\right]_0^{\frac{6-x}{2}} dx$$

$x + 2y + z = 6$

$z = 6 - x - 2y$

In xy plane, $z = 0$

So $6 - x - 2y = 0$

$2y = 6 - x$

$y = \frac{6-x}{2}$

$x = 6$

$$= \int_0^6 6\left(\frac{6-x}{2}\right) - x\left(\frac{6-x}{2}\right) - \left(\frac{6-x}{2}\right)^2 dx$$

$$= \int_0^6 18 - 3x - 3x + \frac{x^2}{2} - \left(\frac{36-12x+x^2}{4}\right) dx$$

$$= \int_0^6 \left(18 - 6x + \frac{x^2}{2} - 9 + 3x - \frac{x^2}{4}\right) dx$$

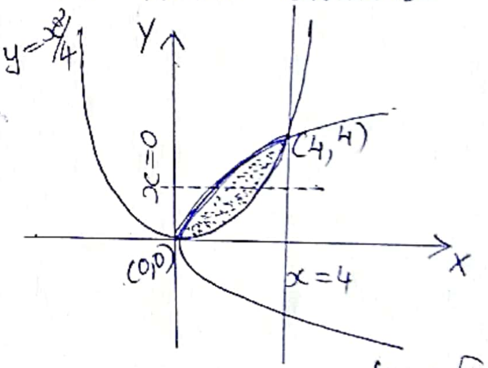
$$= \int_0^6 \left(9 - 3x + \frac{x^2}{4}\right) dx$$

$$= \left[9x - \frac{3x^2}{2} + \frac{1}{4}\frac{x^3}{3}\right]_0^6$$

$$= 54 - 54 + 18 \quad \underline{\underline{= 18 \text{ cu.unit}}}$$

16 a) Change the order of integration and hence evaluate

$$\int_0^4 \int_{\frac{x^2}{4}}^{2\sqrt{x}} dy\,dx.$$



y limit : $y = \frac{x^2}{4}$, $y = 2\sqrt{x}$

x limit : $x = 0$, $x = 4$

$$\int_0^4 \int_{\frac{x^2}{4}}^{2\sqrt{x}} dy\,dx = \int_0^4 \int_{y^2/4}^{2\sqrt{y}} dx\,dy$$

$$= \int_0^4 [x]_{y^2/4}^{2\sqrt{y}} dy$$

$$= \int_0^4 \left(2\sqrt{y} - \frac{y^2}{4}\right) dy$$

$y = \frac{x^2}{4}$, $y = 2\sqrt{x}$

$\frac{x^2}{4} = 2\sqrt{x}$

$x^2 = 8\sqrt{x}$

$x^4 = 64x$

$x^3 = 64$

$x = 4$

$$= \int_0^4 \left(2y^{1/2} - \frac{y^2}{4}\right) dy$$

$$= \left[\frac{2 \cdot y^{3/2}}{3/2} - \frac{1}{4}\frac{y^3}{3}\right]_0^4 = \left[\frac{4}{3} y^{3/2} - \frac{1}{12} y^3\right]_0^4$$

$$= \frac{4}{3}(4)^{3/2} - \frac{1}{12} \cdot 4^3 = \frac{4}{3}(2^2)^{3/2} - \frac{64}{12}$$

$$= \frac{4}{3} \times 8 - \frac{16}{3} = \underline{\underline{\frac{16}{3}}}$$

b) Evaluate $\iiint_G z\, dV$, where G is the wedge in the first octant cut off from the cylindrical solid $y^2 + z^2 \leq 1$ and the planes $y = x$ and $x = 0$.

cylinder $y^2 + z^2 = 1$

$z^2 = 1 - y^2$

$z = \pm\sqrt{1-y^2}$

1^{st} octant, so

$z = \sqrt{1-y^2}$

$y^2 = 1$ (let $z = 0$

$y = \pm 1$

$$\iiint_G z\, dV = \iiint z\, dz\, dx\, dy$$

$$= \int_0^1 \int_0^y \int_0^{\sqrt{1-y^2}} z\, dz\, dx\, dy$$

$$= \int_0^1 \int_0^y \left[\frac{z^2}{2}\right]_0^{\sqrt{1-y^2}} dx\, dy$$

$$= \frac{1}{2} \int_0^1 \int_0^y (1-y^2)\, dx\, dy$$

$$= \frac{1}{2} \int_0^1 (1-y^2)\, dy \, [x]_0^y = \frac{1}{2} \int_0^1 (1-y^2)\, y\, dy$$

$$= \frac{1}{2} \int_0^1 (y - y^3)\, dy = \frac{1}{2}\left[\frac{y^2}{2} - \frac{y^4}{4}\right]_0^1$$

$$= \frac{1}{2}\left(\frac{1}{2} - \frac{1}{4}\right) = \frac{1}{8} //$$

17

Module-4

a) Test the convergence of the series $1 + \frac{1.3}{3!} + \frac{1.3.5}{5!} + \cdots$

$$1 + \frac{1.3}{3!} + \frac{1.3.5}{5!} + \frac{1.3.5.7}{7!} + \cdots = \sum_{n=1}^{\infty} \frac{1.3.5\cdots(2n-1)}{(2n-1)!}$$

$$u_n = \frac{1.3.5.\cdots(2n-1)}{(2n-1)!}$$

$$u_{n+1} = \frac{1.3.5.\cdots(2n-1)(2n+1)}{(2n+1)!}$$

$$\frac{u_{n+1}}{u_n} = \frac{\cancel{1.3.5\cdots(2n-1)}(2n+1)}{(2n+1)!} \times \frac{(2n-1)!}{\cancel{1.3.5.7\cdots(2n-1)}}$$

$$= \frac{(2n+1)}{2n(2n+1)(2n-1)!} \times (2n-1)! = \frac{1}{2n}$$

$$\lim_{n\to\infty} \frac{u_{n+1}}{u_n} = \lim_{n\to\infty} \frac{1}{2n} = 0 < 1$$

$\therefore$ By Ratio test, the given series converges.

b) Find the sum of the series $\sum_{k=1}^{\infty} \frac{1}{k(k+1)}$

$$u_k = \frac{1}{k(k+1)} = \frac{A}{k} + \frac{B}{k+1}$$

$$1 = A(k+1) + Bk$$

when $k=0$, $A=1$

when $k=-1$, $B=-1$

So $u_k = \frac{1}{k} - \frac{1}{k+1}$

$$\Sigma u_k = \sum_{k=1}^{\infty} \frac{1}{k(k+1)} = \sum_{k=1}^{\infty} \frac{1}{k} - \frac{1}{k+1}$$

$$= \left(1-\frac{1}{2}\right)+\left(\frac{1}{2}-\frac{1}{3}\right)+\left(\frac{1}{3}-\frac{1}{4}\right)+\left(\frac{1}{4}-\frac{1}{5}\right)+\cdots$$

$$= 1-\frac{1}{2}+\frac{1}{2}-\frac{1}{3}+\frac{1}{3}-\frac{1}{4}+\frac{1}{4}\cdots$$

$$= 1, \text{ as } k\to\infty.$$

$$\therefore \sum_{k=1}^{\infty} \frac{1}{k(k+1)} = 1$$

18 a) Test the convergence of (i) $\sum_{k=1}^{\infty} \frac{k!}{3!(k-1)!3^k}$

(ii) $\sum_{k=1}^{\infty} \left(\frac{4k-5}{2k+1}\right)^k$

(i) $u_k = \frac{k!}{3!(k-1)!3^k}$, $u_{k+1} = \frac{(k+1)!}{3!\,k!\,3^{k+1}}$

$$\frac{u_{k+1}}{u_k} = \frac{(k+1)!}{3!\,k!\,3^{k+1}} \times \frac{3!\,(k-1)!\,3^k}{k!}$$

$$= \frac{(k+1)\cancel{k!}}{\cancel{3!}\cancel{k!}\,\cancel{3^k}\cdot 3} \times \frac{\cancel{3!}\,\cancel{(k-1)!}\,\cancel{3^k}}{k\,\cancel{(k-1)!}}$$

$$= \frac{(k+1)}{3k} = \frac{k\left(1+\frac{1}{k}\right)}{3k} = \frac{1+1/k}{3}$$

$$\lim_{k\to\infty} \frac{u_{k+1}}{u_k} = \lim_{k\to\infty} \frac{1+1/k}{3} = \frac{1}{3} < 1$$

$\therefore$ By Ratio test, the given series converges.

(ii) $u_k = \left(\frac{4k-5}{2k+1}\right)^k$

$$(u_k)^{1/k} = \frac{4k-5}{2k+1} = \frac{k(4-5/k)}{k(2+1/k)} = \frac{4-5/k}{2+1/k}$$

$$\lim_{k\to\infty} (u_k)^{1/k} = \lim_{k\to\infty} \frac{4-\frac{5}{k}}{2+\frac{1}{k}} = 2 > 1$$

$\therefore$ By Root test, the given series diverges.

b) Test the absolute or conditional convergence of

$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1} k^2}{k^3+1}.$$

$$\sum u_k = \sum_{k=1}^{\infty} \frac{(-1)^{k+1} k^2}{k^3+1}$$

$$\sum |u_k| = \sum_{k=1}^{\infty} \frac{k^2}{k^3+1} = \sum_{k=1}^{\infty} \frac{k^2}{k^3(1+\frac{1}{k^3})}$$

$\sum V_k = \sum \frac{1}{k}$, diverges by p-series test.

$\lim_{k\to\infty} \frac{u_k}{V_k} = \lim_{k\to\infty} \frac{1}{1+\frac{1}{k^3}} = 1$, finite non zero numbers.

$\therefore$ By comparison test, $\sum |u_k|$ is divergent.

Hence $\sum u_k$ is not absolutely convergent.

By Leibnitz test,

$$u_k = \frac{k^2}{k^3+1}, \quad u_{k+1} = \frac{(k+1)^2}{(k+1)^3+1}$$

$$u_k - u_{k+1} = \frac{k^2}{k^3+1} - \frac{(k+1)^2}{(k+1)^3+1}$$

$$= \frac{k^2[(k+1)^3+1] - (k^3+1)(k+1)^2}{(k^3+1)[(k+1)^3+1]}$$

$$= \frac{k^4+2k^3+k^2-2k-1}{(k^3+1)((k+1)^3+1)} > 0$$

$\therefore u_k > u_{k+1}$

$$\lim_{k\to\infty} u_k = \lim_{k\to\infty} \frac{k^2}{k^3+1} = \lim_{k\to\infty} \frac{k^2}{k^3(1+\frac{1}{k^3})}$$

$$= \lim_{k\to\infty} \frac{1}{k(1+\frac{1}{k^3})} = 0$$

$\therefore$ By Leibnitz test, Σu_k converges.

Hence the given series converges conditionally.

Module 5

19

a) Expand into a Fourier series, $f(x) = e^{-x}$, $0 < x < 2\pi$

$$l = \frac{2\pi - 0}{2} = \pi$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$$

$$a_0 = \frac{1}{l}\int f(x)\,dx = \frac{1}{\pi}\int_0^{2\pi} e^{-x}\,dx$$

$$= \frac{1}{\pi}\left[\frac{e^{-x}}{-1}\right]_0^{2\pi} = \frac{-1}{\pi}\left[e^{-2\pi} - e^{0}\right] = \frac{1-e^{-2\pi}}{\pi}$$

$$a_n = \frac{1}{l}\int f(x)\cos\left(\frac{n\pi x}{l}\right)dx$$

$$= \frac{1}{\pi}\int_0^{2\pi} e^{-x}\cos nx\,dx$$

$$= \frac{1}{\pi}\left[\frac{e^{-x}}{1+n^2}(-\cos nx + n\sin nx)\right]_0^{2\pi}$$

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$$= \frac{1}{\pi}\left[\frac{e^{-2\pi}}{1+n^2}(-\cos 2n\pi + n\sin 2n\pi) - \frac{e^{0}}{1+n^2}(-\cos 0 + n\sin 0)\right]$$

$$a_n = \frac{1}{\pi(1+n^2)}\left(1-e^{-2\pi}\right)$$

$$b_n = \frac{1}{l}\int f(x)\sin\left(\frac{n\pi x}{l}\right)dx$$

$$= \frac{1}{\pi}\int_0^{2\pi} e^{-x}\sin nx\,dx$$

$$= \frac{1}{\pi}\left[\frac{e^{-x}}{1+n^2}(-\sin nx - n\cos nx)\right]_0^{2\pi}$$

$$= \frac{1}{\pi}\left[\frac{e^{-2\pi}}{1+n^2}(-\sin 2n\pi - n\cos 2n\pi) - \frac{e^{0}}{1+n^2}(-\sin 0 - n\cos 0)\right]$$

$$= \frac{n}{\pi(1+n^2)}\left(1-e^{-2\pi}\right)$$

$$\therefore e^{-x} = \frac{1-e^{-2\pi}}{2\pi} + \frac{(1-e^{-2\pi})}{\pi}\sum_{n=1}^{\infty}\frac{\cos nx + n\sin nx}{1+n^2}$$

---

b) Find the half range cosine series for $f(x)=(x-1)^2$ in $0 \le x \le 1$

$l = 1-0 = 1$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right)$$

$$a_0 = \frac{2}{l}\int f(x)dx = 2\int_0^1 (x-1)^2 dx = 2\left[\frac{(x-1)^3}{3}\right]_0^1$$

$$= \frac{2}{3}\left[0 - -1\right] = 2/3$$

$$a_n = \frac{2}{l}\int f(x)\cos\left(\frac{n\pi x}{l}\right)dx$$

$$= 2\int_0^1 (x-1)^2 \cdot \cos(n\pi x)\, dx$$

$u = (x-1)^2$ $\quad\quad$ $v = \cos(n\pi x)$

$u' = 2(x-1)$ $\quad\quad$ $v_1 = \frac{\sin(n\pi x)}{n\pi}$

$u'' = 2$ $\quad\quad$ $v_2 = -\frac{\cos(n\pi x)}{n^2\pi^2}$

$u''' = 0$ $\quad\quad$ $v_3 = -\frac{\sin(n\pi x)}{n^3\pi^3}$

$\int uv = uv_1 - u'v_2 + u''v_3 - + \ldots$

$$\therefore a_n = 2\left[(x-1)^2 \cdot \frac{\sin(n\pi x)}{n\pi} - 2(x-1)\cdot \frac{-\cos(n\pi x)}{n^2\pi^2} + 2\cdot\frac{-\sin(n\pi x)}{n^3\pi^3}\right]_0^1$$

$$= 2\left[0+0-0-\left(0+\frac{-2\cdot\cos n\pi\times 0}{n^2\pi^2}+0\right)\right]$$

$$= 2 + 2\cdot\frac{1}{n^2\pi^2} = \frac{2\left(1+n^2\pi^2\right)}{n^2\pi^2}$$

$\therefore$ Cosine series is

$$(x-1)^2 = \frac{2/3}{2} + \sum_{n=1}^{\infty} \frac{2\left(1+n^2\pi^2\right)}{n^2\pi^2}\cos(n\pi x)$$

$$= \frac{1}{3} + \frac{2}{\pi^2}\sum_{n=1}^{\infty}\left(\frac{1+n^2\pi^2}{n^2}\right)\cos(n\pi x)$$

20

a) Find the Fourier series of the function $f(x) = |x|$ in $-1 \le x \le 1$.

$$l = \frac{1-(-1)}{2} = 1$$

$|x|$ is an even function

$$a_0 = \frac{1}{l}\int f(x)\,dx = \int_{-1}^{1} |x|\,dx$$

$$= 2\int_0^1 x\,dx = 2\left[\frac{x^2}{2}\right]_0^1$$

$$a_0 = \underline{\underline{1}}$$

$$a_n = \frac{1}{l}\int f(x)\cos\left(\frac{n\pi x}{l}\right)dx = \int_{-1}^{1} |x| \cos(n\pi x)\,dx$$

$$= 2\int_0^1 x\cos(n\pi x)\,dx$$

$u = x \qquad v = \cos(n\pi x)$

$u' = 1 \qquad v_1 = \dfrac{\sin(n\pi x)}{n\pi}$

$\qquad\qquad v_2 = -\dfrac{\cos(n\pi x)}{n^2\pi^2}$

$$\int uv = uv_1 - u'v_2 + \cdots$$

$$\therefore a_n = 2\left[x \cdot \frac{\sin n\pi x}{n\pi} - 1 \cdot -\frac{\cos(n\pi x)}{n^2\pi^2}\right]_0^1$$

$$= 2\left[0 + \frac{\cos n\pi}{n^2\pi^2} - \left(0 + \frac{1}{n^2\pi^2}\right)\right]$$

$$= 2\left[\frac{(-1)^n}{n^2\pi^2} - \frac{1}{n^2\pi^2}\right] = \underline{\underline{\frac{2}{n^2\pi^2}}}\left((-1)^n - 1\right)$$

$$b_n = \frac{1}{l}\int f(x)\sin\left(\frac{n\pi x}{l}\right)dx = 0$$

$$\therefore f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n\cos\left(\frac{n\pi x}{l}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$$

$$= \frac{1}{2} + \sum_{n=1}^{\infty} \frac{2}{n^2\pi^2}\left[(-1)^n - 1\right]\cos(n\pi x)$$

D) Find the Fourier Sine Series of $f(x) = x\cos x$ in $0 < x < \pi$.

Fourier Sine series is

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$$

$2\cos A \sin B$
$= \sin(A+B)$
$- \sin(A-B)$

$l = \pi$

$$b_n = \frac{2}{l}\int_0^{\pi} f(x)\cdot \sin\left(\frac{n\pi x}{l}\right)dx$$

$$= \frac{2}{\pi}\int_0^{\pi} x\cos x \cdot \sin nx \, dx = \frac{1}{\pi}\int_0^{\pi} x \cdot 2\cos x \sin nx \, dx$$

$$= \frac{2 1}{\pi}\int_0^{\pi} x\left[\sin(x+nx) - \sin(x-nx)\right]dx$$

$$= \frac{1}{\pi}\left[\int_0^{\pi} x\sin(1+n)x \, dx - \int_0^{\pi} x\sin(1-n)x \, dx\right]$$

$$= \frac{1}{\pi}\left\{\left[x\cdot\frac{-\cos(1+n)x}{1+n} - 1\cdot\frac{-\sin(1+n)x}{(1+n)^2}\right]_0^{\pi} - \left[x\cdot\frac{-\cos(1-n)x}{1-n} - 1\cdot\frac{-\sin(1-n)x}{(1-n)^2}\right]_0^{\pi}\right\}$$

$$= \frac{1}{\pi}\left[-\frac{\pi}{1+n}\cos(1+n)\pi + 0 - \left(-\frac{\pi}{1-n}\cos(1-n)\pi + 0\right) - (0+0-0+0)\right]$$

$$= \frac{1}{\pi}\left[-\frac{\pi}{1+n}\cos(\pi+n\pi) + \frac{\pi}{1-n}\cos(\pi-n\pi)\right]$$

$$= \frac{1}{\pi}\times\pi\left[\frac{+\cos n\pi}{1+n} - \frac{\cos n\pi}{1-n}\right] = \cos n\pi\left(\frac{1}{1+n} - \frac{1}{1-n}\right)$$

$$= (-1)^n\left[\frac{1-n-(1+n)}{1-n^2}\right] = (-1)^n\cdot\frac{-2n}{1-n^2} = \frac{(-1)^{n+1}2n}{1-n^2}$$

$$\therefore f(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}\, 2n}{(1-n^2)} \sin nx$$

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